Six full geometry groups from an intrinsic 352-object relation

Let $G = \mathrm{HS}$ in the acquired faithful permutation model. We retain the complete coset catalogue, the literal full $\mathrm{Aut}(G)$ action, and the reviewed source coordinates for D. Leemans, *The residually weakly primitive geometries of HS*, Australasian Journal of Combinatorics 33 (2005), 231–236, with its archived supplementary diagrams.

Theorem 1. *For the following six families the full type-preserving automorphism group is G. Their full correlation groups, allowing type permutations, are as follows. Here B is the base chamber stabilizer, and S, T are the ordered and setwise parabolic-family stabilizers in* $\mathrm{Aut}(G)$.

Family	*Source*	$\lvert B\rvert$	$\lvert S\rvert$	$\lvert T\rvert$	$\mathrm{Cor}(\Gamma)$
21	*3.13*	*72*	*72*	*72*	G
22	*3.16*	*72*	*72*	*144*	$G \times C_2$
37	*4.31*	*4*	*4*	*8*	$G \times C_2$
39	*4.29*	*4*	*4*	*4*	G
54	*4.14*	*8*	*8*	*8*	G
56	*4.18*	*8*	*8*	*16*	$G \times C_2$

The proof uses explicit finite maps and exact group operations, with the complete coloured-graph algorithm as a stated computational premise.

A smaller intrinsic structure. Every listed family has exactly one type a of cardinality $n = 352$. It is therefore fixed setwise by every correlation. Let b be its base object, $P_a = G_b$, and let O_k be the P_a-orbits on that type, of sizes d_k. For each type j, the point shadow of its base object is $S_j = b^{P_j}$. If e_{jk} counts the ordered distinct pairs in S_j belonging to pair orbit k, then the number of type-j objects incident with both entries of such a pair is

$$m_{jk} = \frac{[G : P_j]e_{jk}}{352d_k}.$$

To prove this, count pairs consisting of an object and an ordered pair in its shadow. The total is $[G : P_j]e_{jk}$; the pair orbit has $352d_k$ elements. Transitivity makes the fibre size constant. Objects with equal shadows are still counted with their proper multiplicity. For $j = a$, equality is the within-type shadow convention, so all off-diagonal counts vanish.

The sorted multiset of $([G : P_j], m_{jk})$ over all types is invariant under every correlation. Its distinct values therefore define intrinsic relations on ordered distinct points. In all six literal models the point-stabilizer subdegrees are $1, 175, 175, 1$; the non-diagonal signatures define two relations of valencies 176 and 175. The unique valency 175 relation is checked to be symmetric. It gives a graph on just 352 vertices. This construction never requires a transversal or permutation action on any of the larger object types.

Literal coordinates and faithfulness. The recorded generator images give a bijective homomorphism from the actual acquired G to a permutation group E on 352 points. The base-point stabilizer is checked to be the actual image of P_a. All saved elements $t_x \in G$ send the base point to the corresponding point x. These checks establish the coset identification $P_a g \mapsto b^g$,

without any assumption about historical labels. For a pair (x, y), its orbital is determined by the E_b-orbit of $y^{t_x^{-1}}$. This computes every e_{jk} from the saved actual shadows and reconstructs every graph row by equivariance.

A point shadow determines a type-j object exactly when its setwise stabilizer in E is the image of P_j. All types in families $21, 22, 37, 54, 56$ meet this condition directly. In family 39, the first type initially has shadow stabilizer of order 1440, whereas its parabolic has order 32. The other three types are already identified. Adjoin the set of encodings of its type-3 neighbours. This is the actual orbit of the base type-3 encoding under P_1; its size is $[P_1 : P_1 \cap P_3] = 2$. Its stabilizer inside the old stabilizer has order 32 and equals the actual image of P_1. The saved certificate also records the preceding redundant refinement by type 2, which leaves order 1440; both entries are checked in dependency order, including their complete neighbour orbits.

Thus any typed automorphism fixing the 352 objects pointwise fixes every object. This is the acquired injective-encoding lemma: the base encoding has stabilizer exactly its parabolic, and induction through previously identified types gives the same implication for all successive neighbour descriptions.

Characteristic group and full correlations. A complete nauty calculation gives graph-group order 88 704 000 in every case. Every returned generator is checked on every adjacency row. For their literal permutation group D, GAP checks

$$|D| = 88\,704\,000, \qquad E \leq D, \qquad D' = E.$$

In particular E is characteristic in the full graph group. If c is any correlation and $g \in G$, the action of cgc^{-1} on the intrinsic point type therefore agrees with some $h \in G$. The typed automorphism $h^{-1}cgc^{-1}$ fixes that type pointwise, so the preceding faithfulness argument makes it trivial. Hence all full correlations normalize the specified G.

The acquired chamber-stabilizer argument now gives

$$|\operatorname{Aut}(\Gamma)| = |G : B||S|, \qquad |\operatorname{Cor}(\Gamma)| = |G : B||T|.$$

Complete successive subgroup transporters test every possible type permutation. Every positive witness is checked on the entire actual parabolic family; impossible order or intersection-order patterns are only rejection criteria. The resulting S, T are those in the table, with $S \cap G = B$. Therefore every typed automorphism lies in G. The centralizer of G in the full correlation group is the checked quotient $(T \cap G)/B$. It is trivial for $21, 39, 54$, and has order two for $22, 37, 56$. In the latter cases it is central, intersects the centreless G trivially, and together with G has the full correlation order. This proves the direct products in the theorem.

Attribution and boundary. The common-incidence formula, shadow lemma and normalizer argument are acquired theory; this is their finite application at degree 352. Exact GAP arithmetic and nauty completeness remain computational premises. The preceding failed joint-action attempts are preserved; this changed reduction repeats no ambient subgroup or geometry census. The source-coordinate map is retained separately from structural names. No abstract Borel-name identification, whole-source completion, historical novelty, question resolution, Lean or measurement claim is supplied by this component.